

Лабораторная работа №3

Настройка DHCP-сервера

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1. Цели и задачи работы
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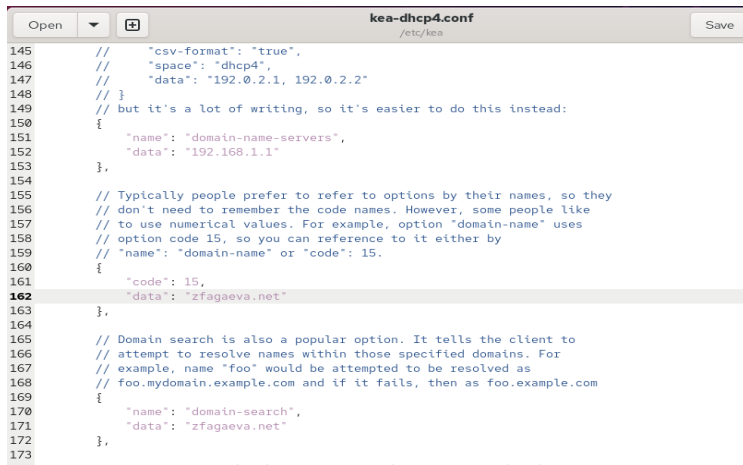
1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Получить практические навыки установки, настройки и диагностики DHCP-сервера Kea на виртуальной машине **server**.

2. 2. Процесс выполнения лабораторной работы

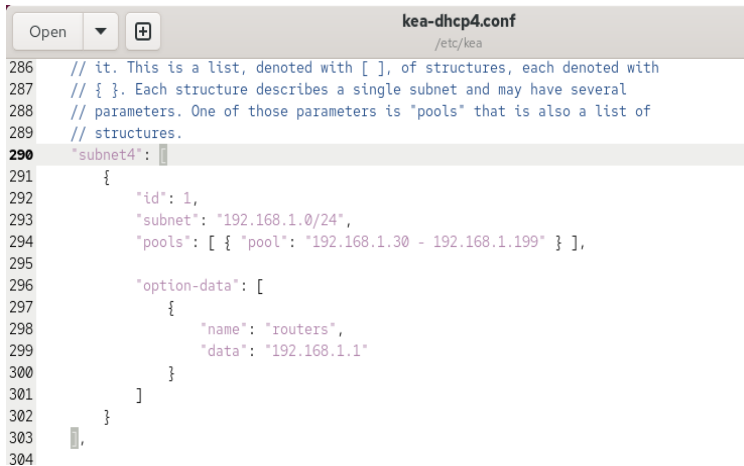
2.1 Настраиваю параметры DNS и доменной зоны



```
145 //      "csv-format": "true",
146 //      "space": "dhcp4",
147 //      "data": "192.0.2.1, 192.0.2.2"
148 // }
149 // but it's a lot of writing, so it's easier to do this instead:
150 {
151     "name": "domain-name-servers",
152     "data": "192.168.1.1"
153 },
154
155 // Typically people prefer to refer to options by their names, so they
156 // don't need to remember the code names. However, some people like
157 // to use numerical values. For example, option "domain-name" uses
158 // option code 15, so you can reference to it either by
159 // "name": "domain-name" or "code": 15.
160 {
161     "code": 15,
162     "data": "zfagaeva.net"
163 },
164
165 // Domain search is also a popular option. It tells the client to
166 // attempt to resolve names within those specified domains. For
167 // example, name "foo" would be attempted to be resolved as
168 // foo.mydomain.example.com and if it fails, then as foo.example.com
169 {
170     "name": "domain-search",
171     "data": "zfagaeva.net"
172 },
173
174 ...
```

Рисунок 1: Настройка параметров DNS и доменной зоны в конфигурации Kea DHCP

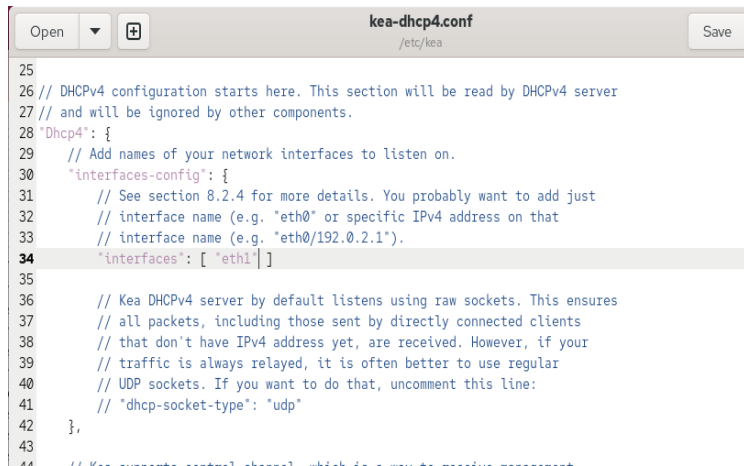
2.2 Настраиваю DHCP-подсеть



```
Open ▼ + kea-dhcp4.conf /etc/kea
286 // it. This is a list, denoted with [ ], of structures, each denoted with
287 // { }. Each structure describes a single subnet and may have several
288 // parameters. One of those parameters is "pools" that is also a list of
289 // structures.
290 "subnet4": [
291     {
292         "id": 1,
293         "subnet": "192.168.1.0/24",
294         "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],
295
296         "option-data": [
297             {
298                 "name": "routers",
299                 "data": "192.168.1.1"
300             }
301         ]
302     }
303 ],
304
```

Рисунок 2: Настройка подсети и диапазона адресов DHCP

2.3 Привязываю DHCP к интерфейсу eth1



```
25
26 // DHCPv4 configuration starts here. This section will be read by DHCPv4 server
27 // and will be ignored by other components.
28 "Dhcp4": {
29     // Add names of your network interfaces to listen on.
30     "interfaces-config": {
31         // See section 8.2.4 for more details. You probably want to add just
32         // interface name (e.g. "eth0" or specific IPv4 address on that
33         // interface name (e.g. "eth0/192.0.2.1").
34         "interfaces": [ "eth1" ]
35
36         // Kea DHCPv4 server by default listens using raw sockets. This ensures
37         // all packets, including those sent by directly connected clients
38         // that don't have IPv4 address yet, are received. However, if your
39         // traffic is always relayed, it is often better to use regular
40         // UDP sockets. If you want to do that, uncomment this line:
41         // "dhcp-socket-type": "udp"
42     },
43
44     // Kea DHCPv4 server by default listens using raw sockets. This ensures
```

Рисунок 3: Привязка Kea DHCP к интерфейсу eth1

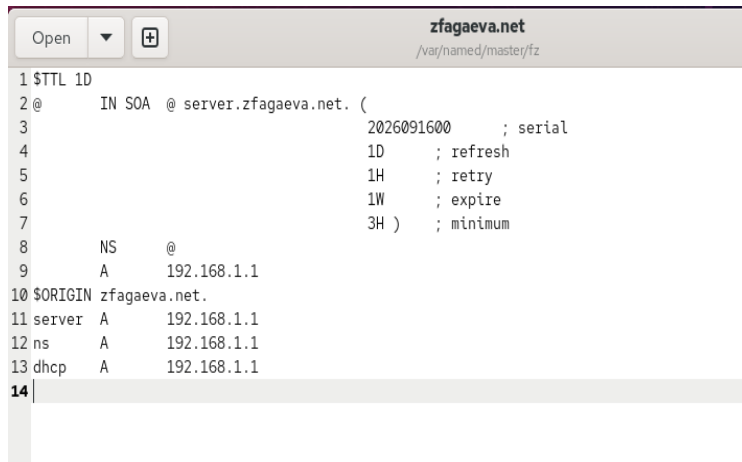
2.4 Проверяю конфигурацию DHCP-сервера

```
Installed:
  kea-3.0.3-1.el10_2.x86_64          kea-libs-3.0.3-1.el10_2.x86_64
  libpq-16.14-1.el10_2.x86_64       log4cplus-2.1.1-10.el10.x86_64
  mariadb-connector-c-3.4.4-2.el10_2.x86_64 mariadb-connector-c-config-3.4.4-2.el10_2.noarch

Complete!
[root@server.zfagaeva.net ~]# cp /etc/kea/kea-dhcp4.conf /etc/kea/kea-dhcp4.conf_$(date +%Y-%m-%d)
[root@server.zfagaeva.net ~]# gedit /etc/kea/kea-dhcp4.conf
[root@server.zfagaeva.net ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-16 08:53:19.882 WARN [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_RESERVED_QUEUE_CONTROL disabling dhcp
queue control when multi-threading is enabled.
2026-09-16 08:53:19.882 WARN [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-threa
ding is enabled and host reservations lookup is always performed first.
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_CFGMGR_NEW_SUBNET4 a new subnet has been
added to configuration: 192.168.1.0/24 with params: t1=900, t2=1800, valid-lifetime=3600
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_CFGMGR_SOCKET_TYPE_SELECT using socket ty
pe raw
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_CFGMGR_ADD_IFACE listening on interface e
th1
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-t
ype" not specified , using default socket type raw
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_LEASE_MGR_BACKENDS_REGISTERED the followi
ng lease backend types are available: memfile
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_HOSTS_BACKENDS_REGISTERED the following host backen
d types are available:
2026-09-16 08:53:19.883 INFO [kea-dhcp4.dhcp4/12421.140360134690944] DHCP4_FORENSIC_BACKENDS_REGISTERED the followin
g forensic backend types are available:
2026-09-16 08:53:19.884 INFO [kea-dhcp4.database/12421.140360134690944] CONFIG_BACKENDS_REGISTERED the following config
backend types are available:
[root@server.zfagaeva.net ~]#
```

Рисунок 4: Установка Кеа и проверка конфигурации DHCP-сервера

2.5 Добавляю DHCP-сервер в прямую DNS-зону

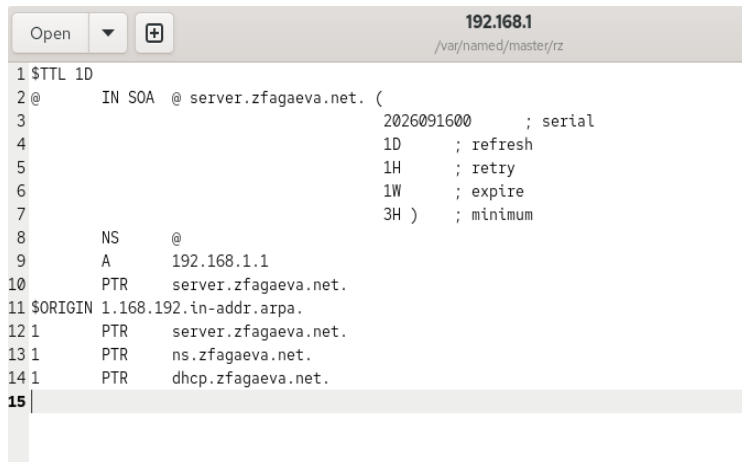


The screenshot shows a DNS zone editor interface for the domain **zfagaeva.net**. The file path is `/var/named/master/fz`. The editor contains a list of records with line numbers 1 through 14. The records are as follows:

```
1 $TTL 1D
2 @      IN SOA  @ server.zfagaeva.net. (
3                               2026091600      ; serial
4                               1D      ; refresh
5                               1H      ; retry
6                               1W      ; expire
7                               3H )    ; minimum
8      NS      @
9      A      192.168.1.1
10 $ORIGIN zfagaeva.net.
11 server A     192.168.1.1
12 ns     A     192.168.1.1
13 dhcp   A     192.168.1.1
14 |
```

Рисунок 5: Добавление DHCP-сервера в прямую DNS-зону zfagaeva.net

2.6 Добавляю DHCP-сервер в обратную DNS-зону



```
1 $TTL 1D
2 @      IN SOA  @ server.zfagaeva.net. (
3                               2026091600      ; serial
4                               1D      ; refresh
5                               1H      ; retry
6                               1W      ; expire
7                               3H )    ; minimum
8      NS      @
9      A      192.168.1.1
10     PTR     server.zfagaeva.net.
11 $ORIGIN 1.168.192.in-addr.arpa.
12 1      PTR     server.zfagaeva.net.
13 1      PTR     ns.zfagaeva.net.
14 1      PTR     dhcp.zfagaeva.net.
15
```

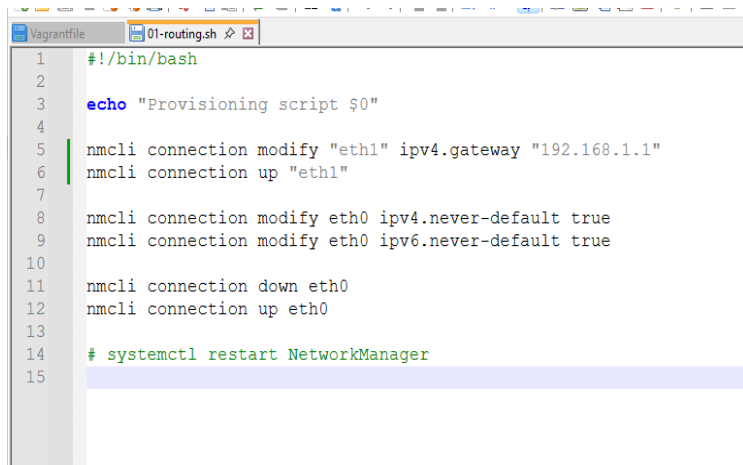
Рисунок 6: Добавление DHCP-сервера в обратную DNS-зону

2.7 Проверяю доступность DHCP-сервера

```
[root@server.zfagaeva.net ~]#  
[root@server.zfagaeva.net ~]# systemctl restart named  
[root@server.zfagaeva.net ~]# ping dhcp.zfagaeva.net  
PING dhcp.zfagaeva.net (192.168.1.1) 56(84) bytes of data.  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.018 ms  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.045 ms  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.045 ms  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.037 ms  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=5 ttl=64 time=0.042 ms  
64 bytes from server.zfagaeva.net (192.168.1.1): icmp_seq=6 ttl=64 time=0.044 ms  
^C  
--- dhcp.zfagaeva.net ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 513ms  
rtt min/avg/max/mdev = 0.018/0.038/0.045/0.009 ms  
[root@server.zfagaeva.net ~]# firewall-cmd --add-service=dhcp  
success  
[root@server.zfagaeva.net ~]# firewall-cmd --add-service=dhcp --permanent  
success  
[root@server.zfagaeva.net ~]# restorecon -vR /etc  
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0  
[root@server.zfagaeva.net ~]# restorecon -vR /var/named/  
[root@server.zfagaeva.net ~]# restorecon -vR /var/lib/kea/  
[root@server.zfagaeva.net ~]# systemctl start kea-dhcp4.service  
[root@server.zfagaeva.net ~]# █
```

Рисунок 7: Проверка DNS-имени DHCP-сервера, настройка firewall и запуск Kea DHCP

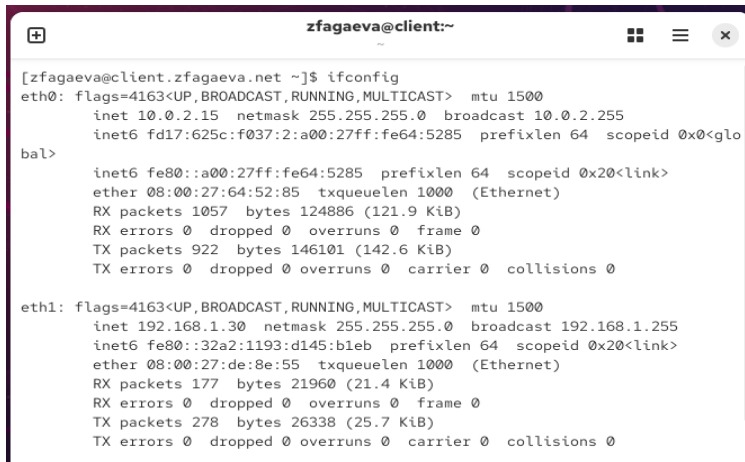
2.8 Настраиваю маршрутизацию клиента



```
Vagrantfile 01-routing.sh x
1  #!/bin/bash
2
3  echo "Provisioning script $0"
4
5  nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
6  nmcli connection up "eth1"
7
8  nmcli connection modify eth0 ipv4.never-default true
9  nmcli connection modify eth0 ipv6.never-default true
10
11 nmcli connection down eth0
12 nmcli connection up eth0
13
14 # systemctl restart NetworkManager
15
```

Рисунок 8: Скрипт настройки маршрутизации виртуальной машины client

2.9 Проверяю сетевые интерфейсы клиента



```
zfagaeva@client:~  
[zfagaeva@client.zfagaeva.net ~]$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fd17:625c:f037:2:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x0<global>  
    ether 08:00:27:64:52:85 txqueuelen 1000 (Ethernet)  
    RX packets 1057 bytes 124886 (121.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 922 bytes 146101 (142.6 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.1.30 netmask 255.255.255.0 broadcast 192.168.1.255  
    inet6 fe80::32a2:1193:d145:b1eb prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:de:8e:55 txqueuelen 1000 (Ethernet)  
    RX packets 177 bytes 21960 (21.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 278 bytes 26338 (25.7 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рисунок 9: Анализ сетевых интерфейсов виртуальной машины client

2.10 Анализирую выданную DHCP-аренду

```
[root@server.zfagaeva.net ~]# restorecon -vR /var/lib/kea/
[root@server.zfagaeva.net ~]# systemctl start kea-dhcp4.service
[root@server.zfagaeva.net ~]#
[root@server.zfagaeva.net ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
192.168.1.30,08:00:27:de:8e:55,01:08:00:27:de:8e:55,3600,1789553082,1,0,0,client,0,,0
[root@server.zfagaeva.net ~]#
```

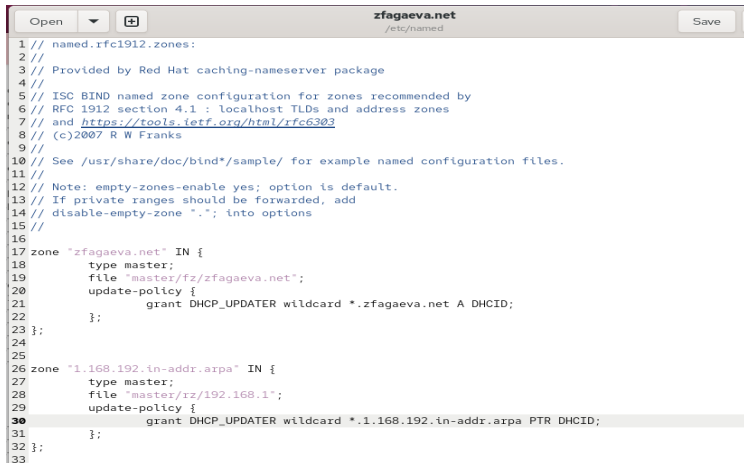
Рисунок 10: Просмотр выданной DHCP-аренды в файле kea-leases4.csv

2.11 Создаю TSIG-ключ

```
[root@server.zfagaeva.net ~]#  
[root@server.zfagaeva.net ~]# mkdir -p /etc/named/keys  
[root@server.zfagaeva.net ~]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key  
[root@server.zfagaeva.net ~]# cat /etc/named/keys/dhcp_updater.key  
key "DHCP_UPDATER" {  
    algorithm hmac-sha512;  
    secret "0isahpX8u+j84UAIcugtnCZcWYVntW7ierySGeNrI0PEuu5HZIpc8MHCrJ7rgd/BYtxn5QmFp2L0uo7srjZ0w==";  
};  
[root@server.zfagaeva.net ~]# chown -R named:named /etc/named/keys/  
[root@server.zfagaeva.net ~]#
```

Рисунок 11: Создание TSIG-ключа DHCP_UPDATER для динамического обновления DNS

2.12 Разрешаю динамическое обновление DNS-зон



```
1 // named.rfc1912.zones:
2 //
3 // Provided by Red Hat caching-nameserver package
4 //
5 // ISC BIND named zone configuration for zones recommended by
6 // RFC 1912 section 4.1 : localhost TLDs and address zones
7 // and https://tools.ietf.org/html/rfc6303
8 // (c)2007 R W Franks
9 //
10 // See /usr/share/doc/bind*/sample/ for example named configuration files.
11 //
12 // Note: empty-zones-enable yes; option is default.
13 // If private ranges should be forwarded, add
14 // disable-empty-zone "."; into options
15 //
16
17 zone "zfagaeva.net" IN {
18     type master;
19     file "master/fz/zfagaeva.net";
20     update-policy {
21         grant DHCP_UPDATER wildcard *.zfagaeva.net A DHCID;
22     };
23 };
24
25
26 zone "1.168.192.in-addr.arpa" IN {
27     type master;
28     file "master/rz/192.168.1";
29     update-policy {
30         grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa PTR DHCID;
31     };
32 };
33
```

Рисунок 12: Разрешение динамического обновления прямой и обратной DNS-зон

2.13 Настраиваю ключ для Кеа



```
1 "tsig-keys": [  
2   {  
3     "name": "DHCP_UPDATER",  
4     "algorithm": "hmac-sha512",  
5     "secret": "0isahpX8u+j84UAicugtnCZcWYVntW7ierySGeNrI0PEuu5HZIpc8MHCcrJ7rgd/BYtxn5QmFp2Lowo7srjZ0w=="  
6   }  
7 ]
```

Рисунок 13: Настройка TSIG-ключа в формате Кеа

2.14 Настраиваю DHCP-DDNS



```
16
17 // DHCP DDNS configuration starts here. This is a very simple configuration
18 // that simply starts the DDNS daemon, but will not do anything useful.
19 // See Section 11 for examples and details description.
20 "DhcpDdns":
21 {
22   "ip-address": "127.0.0.1",
23   "port": 53001,
24   "control-socket": {
25     "socket-type": "unix",
26     "socket-name": "kea-ddns-ctrl-socket"
27   },
28   <?include "/etc/kea/tsig-keys.json" ?>
29   "forward-ddns" : {
30     "ddns-domains" : [
31       {
32         "name": "zfagaeva.net.",
33         "key-name": "DHCP_UPDATER",
34         "dns-servers": [
35           { "ip-address": "192.168.1.1" }
36         ]
37       }
38     ],
39     "reverse-ddns" : {
40       "ddns-domains" : [
41         {
42           "name": "1.168.192.in-addr.arpa.",
43           "key-name": "DHCP_UPDATER",
44           "dns-servers": [
45             { "ip-address": "192.168.1.1" }
46           ]
47         }
48       ]
49     }
50   }
```

Рисунок 14: Настройка прямых и обратных DDNS-обновлений в Kea

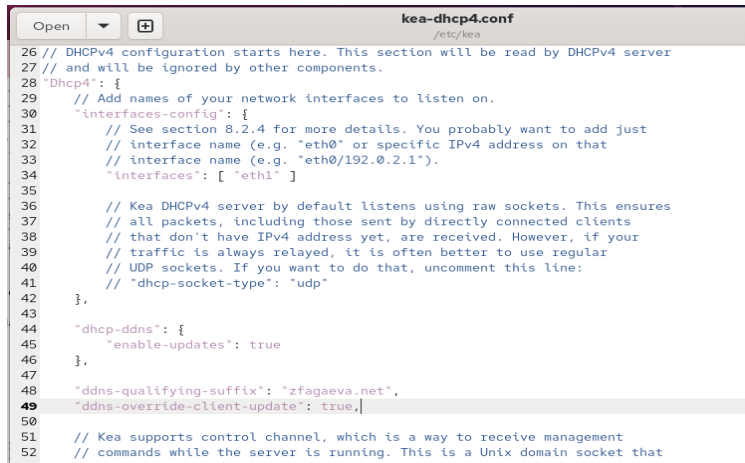
2.15 Запускаю службу Kea DHCP-DDNS

```
[root@server.zfagaeva.net ~]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server.zfagaeva.net ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2026-09-16 09:24:47.693 INFO [kea-dhcp-ddns.dctl/17472.140436734975296] DCTL_CONFIG_CHECK_COMPLETE server has completed
configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), text=Configuration check successf
ul
[root@server.zfagaeva.net ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service' -> '/usr/lib/systemd/system/kea-dhcp-d
dns.service'.
[root@server.zfagaeva.net ~]# systemctl status kea-dhcp-ddns.service
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-16 09:25:11 UTC; 1min 7s ago
 Invocation: ac9c16eb4cf24bc7943d0f963d42dada
    Docs: man:kea-dhcp-ddns(8)
 Main PID: 17713 (kea-dhcp-ddns)
  Status: "Dispatching packets..."
   Tasks: 5 (limit: 10233)
  Memory: 2.3M (peak: 7.1M)
     CPU: 12ms
   CGroup: /system.slice/kea-dhcp-ddns.service
           └─17713 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 16 09:25:11 server.zfagaeva.net systemd[1]: Starting kea-dhcp-ddns.service - Kea DHCP-DDNS Server...
Sep 16 09:25:11 server.zfagaeva.net kea-dhcp-ddns[17713]: 2026-09-16 09:25:11.507 INFO [kea-dhcp-ddns.dctl/17713.1406746
Sep 16 09:25:11 server.zfagaeva.net kea-dhcp-ddns[17713]: INFO COMMAND_ACCEPTOR_START Starting to accept connections via
Sep 16 09:25:11 server.zfagaeva.net kea-dhcp-ddns[17713]: INFO DCTL_CONFIG_COMPLETE server has completed configuration:
Sep 16 09:25:11 server.zfagaeva.net kea-dhcp-ddns[17713]: INFO DHCP_DDNS_STARTED Kea DHCP-DDNS server version 3.0.3 sta
Sep 16 09:25:11 server.zfagaeva.net systemd[1]: Started kea-dhcp-ddns.service - Kea DHCP-DDNS Server.
[root@server.zfagaeva.net ~]#
```

Рисунок 15: Проверка конфигурации и запуск службы Kea DHCP-DDNS

2.16 Включаю DDNS в DHCPv4



```
26 // DHCPv4 configuration starts here. This section will be read by DHCPv4 server
27 // and will be ignored by other components.
28 "Dhcp4": {
29     // Add names of your network interfaces to listen on.
30     "interfaces-config": {
31         // See section 8.2.4 for more details. You probably want to add just
32         // interface name (e.g. "eth0" or specific IPv4 address on that
33         // interface name (e.g. "eth0/192.0.2.1").
34         "interfaces": [ "eth1" ]
35
36         // Kea DHCPv4 server by default listens using raw sockets. This ensures
37         // all packets, including those sent by directly connected clients
38         // that don't have IPv4 address yet, are received. However, if your
39         // traffic is always relayed, it is often better to use regular
40         // UDP sockets. If you want to do that, uncomment this line:
41         // "dhcp-socket-type": "udp"
42     },
43
44     "dhcp-ddns": {
45         "enable-updates": true
46     },
47
48     "ddns-qualifying-suffix": "zfagaeva.net",
49     "ddns-override-client-update": true,
50
51     // Kea supports control channel, which is a way to receive management
52     // commands while the server is running. This is a Unix domain socket that
```

Рисунок 16: Включение динамического обновления DNS в конфигурации DHCPv4

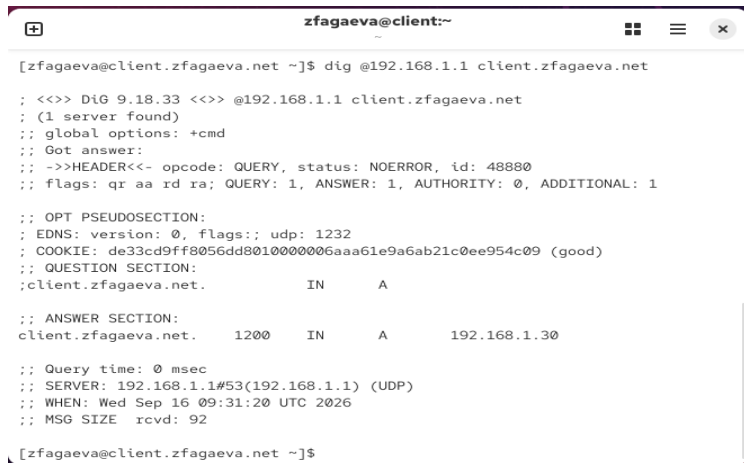
2.17 Проверяю работу DHCPv4

```
backend types are available:
[root@server.zfagaeva.net ~]# systemctl restart kea-dhcp4.service
[root@server.zfagaeva.net ~]# systemctl status kea-dhcp4.service
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-16 09:29:19 UTC; 11s ago
 Invocation: 607ef22cd7284bcb910bc868e1e8fa4f
    Docs: man:kea-dhcp4(8)
   Main PID: 18372 (kea-dhcp4)
   Status: "Dispatching packets..."
    Tasks: 7 (limit: 10233)
  Memory: 2.5M (peak: 7M)
     CPU: 13ms
   CGroup: /system.slice/kea-dhcp4.service
           └─18372 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Sep 16 09:29:19 server.zfagaeva.net systemd[1]: Starting kea-dhcp4.service - Kea DHCPv4 Server...
Sep 16 09:29:19 server.zfagaeva.net kea-dhcp4[18372]: 2026-09-16 09:29:19.581 INFO [kea-dhcp4.dhcp4/18372.1402084889376>
Sep 16 09:29:19 server.zfagaeva.net kea-dhcp4[18372]: 2026-09-16 09:29:19.581 INFO [kea-dhcp4.commands/18372.1402084889>
Sep 16 09:29:19 server.zfagaeva.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
[root@server.zfagaeva.net ~]#
```

Рисунок 17: Проверка работы DHCPv4 после включения DDNS

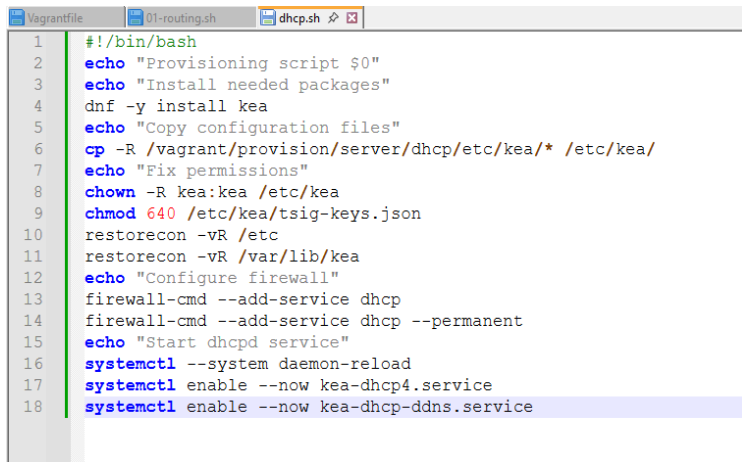
2.18 Проверяю DNS-запись клиента



```
zfagaeva@client:~  
[zfagaeva@client.zfagaeva.net ~]$ dig @192.168.1.1 client.zfagaeva.net  
  
; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.zfagaeva.net  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48880  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: de33cd9ff8056dd80100000006aaa61e9a6ab21c0ee954c09 (good)  
;; QUESTION SECTION:  
;client.zfagaeva.net.                IN      A  
  
;; ANSWER SECTION:  
client.zfagaeva.net.    1200    IN      A      192.168.1.30  
  
;; Query time: 0 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)  
;; WHEN: Wed Sep 16 09:31:20 UTC 2026  
;; MSG SIZE rcvd: 92  
  
[zfagaeva@client.zfagaeva.net ~]$
```

Рисунок 18: Проверка автоматически созданной DNS-записи клиента с помощью dig

2.19 Создаю скрипт dhcp.sh



```
Vagrantfile 01-routing.sh dhcp.sh x
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install kea
5  echo "Copy configuration files"
6  cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
7  echo "Fix permissions"
8  chown -R kea:kea /etc/kea
9  chmod 640 /etc/kea/tsig-keys.json
10 restorecon -vR /etc
11 restorecon -vR /var/lib/kea
12 echo "Configure firewall"
13 firewall-cmd --add-service dhcp
14 firewall-cmd --add-service dhcp --permanent
15 echo "Start dhcpd service"
16 systemctl --system daemon-reload
17 systemctl enable --now kea-dhcp4.service
18 systemctl enable --now kea-dhcp-ddns.service
```

Рисунок 19: Provisioning-скрипт установки и настройки DHCP-сервера

3. 3. Выводы по проделанной работе

3.1 Вывод

В ходе лабораторной работы был установлен и настроен DHCP-сервер Kea на виртуальной машине **server**.

Для внутренней сети **192.168.1.0/24** был задан диапазон адресов **192.168.1.30–192.168.1.199**.

Клиентская машина **client** получила адрес **192.168.1.30** через DHCP.

Было настроено динамическое обновление DNS-зоны с помощью DDNS и TSIG-ключа **DHCP_UPDATER**.

Работоспособность DHCP и DDNS была проверена с помощью **ifconfig**, **ping**, **dig** и анализа файла **kea-leases4.csv**.

Также был создан скрипт **dhcp.sh**, автоматизирующий установку и настройку DHCP-сервера.